

Atrial Fibrillation Screening According to Genetic Risk



A Secondary Analysis of the Randomized LOOP Study

Oliver B. Vad, MD, PhD,^{a,b} Søren Z. Diederichsen, MD, PhD,^a Lucas Y. Xing, MD, PhD,^a Christian Paludan-Müller, MD, PhD,^a Laia M. Monfort, MSc,^{a,b} Gustav Ahlberg, MSc, PhD,^b Laura Andreassen, MD, PhD,^a Ketil J. Haugan, MD, PhD,^c Claus Graff, MSc, PhD,^d Søren Højberg, MD, PhD,^e Derk Krieger, MD, PhD,^{f,g} Axel Brandes, MD, DMSc,^{h,i} Lars Køber, MD, DMSc,^a Jesper H. Svendsen, MD, DMSc^{a,j}

ABSTRACT

BACKGROUND Polygenic risk scores (PRS) hold promise in risk stratification and screening for cardiovascular diseases, including atrial fibrillation (AF).

OBJECTIVES This study investigated the efficacy of AF screening for stroke prevention based on a PRS for AF.

METHODS This prespecified post hoc analysis of the randomized LOOP (Atrial Fibrillation Detected by Continuous ECG Monitoring Using Implantable Loop Recorder to Prevent Stroke in High-Risk Individuals) study included 5,656 AF-naïve individuals aged ≥ 70 years with stroke risk factors and available genetic data. The participants were randomized 1:3 for screening with an implantable loop recorder (ILR) vs usual care. Genetic risk of AF was assessed using a PRS for AF (PRS_{AF}). The primary outcome was a composite of stroke and systemic embolism (SE). Interaction between the randomization arm and PRS_{AF} was assessed in cause-specific Cox regressions for the full cohort and across the observed range of polygenic risk using a continuous prediction grid. Secondary analyses included models stratified by PRS level, gene-screening interactions for major bleeding events, and associations between PRS_{AF} and AF burden (≥ 1 episode lasting ≥ 24 hours among participants with ILR-detected AF).

RESULTS Over a median follow-up period of 5.4 years, 969 participants (17.1%) received a diagnosis of AF, 296 (5.2%) had stroke/SE, and 206 (3.6%) had major bleeding. PRS_{AF} was associated with higher rates of AF (HR per SD increase: 1.20; 95% CI: 1.13-1.28; $P < 0.001$). A significant interaction was observed between ILR screening and PRS_{AF} for stroke/SE ($P_{\text{interaction}} = 0.006$). ILR screening was associated with lower rates of stroke/SE in individuals with PRS_{AF} $\geq$ median (HR: 0.65; 95% CI: 0.43-0.97; $P = 0.036$) but not in those with PRS_{AF} $<$ median (HR: 1.06; 95% CI: 0.72-1.57; $P = 0.75$). ILR screening was associated with higher rates of major bleeding at lower levels of PRS_{AF} ($P_{\text{interaction}} = 0.036$), corresponding to a HR of 1.71 (95% CI: 1.12-2.64; $P = 0.011$) in those with PRS_{AF} $<$ median. A 1-SD increase in PRS_{AF} was associated with an OR of 1.35 (95% CI: 1.02-1.78; $P = 0.037$) for having ≥ 1 AF episode lasting ≥ 24 hours.

CONCLUSIONS ILR screening for AF was associated with a significant reduction in stroke/SE in individuals with higher genetic risk of AF but not in those with lower genetic risk. These hypothesis-generating findings indicate that genetic predisposition may aid in selecting individuals who benefit from AF screening. (JACC. 2026;87:143-153) © 2026 by the American College of Cardiology Foundation.



Listen to this manuscript's
audio summary by
Editor-in-Chief
Dr Harlan M. Krumholz on
www.jacc.org/journal/jacc/
podcasts.

From the ^aDepartment of Cardiology, the Heart Centre, Copenhagen University Hospital-Rigshospitalet, Copenhagen, Denmark; ^bDepartment of Biomedical Sciences, University of Copenhagen, Copenhagen, Denmark; ^cDepartment of Cardiology, Zealand University Hospital Roskilde, Roskilde, Denmark; ^dDepartment of Health Science and Technology, Aalborg University, Gistrup, Denmark; ^eDepartment of Cardiology, Copenhagen University Hospital-Bispebjerg and Frederiksberg, Copenhagen, Denmark; ^fMohammed Bin Rashid University of Medicine and Health Sciences, Dubai, United Arab Emirates; ^gDepartment of Neurology, Mediclinic Parkview Hospital, Dubai, United Arab Emirates; ^hDepartment of Regional Health Research, Faculty of Health Sciences, University of Southern Denmark, Esbjerg, Denmark; ⁱDepartment of Cardiology, Esbjerg Hospital-University Hospital of Southern Denmark, Esbjerg, Denmark; and the ^jDepartment of Clinical Medicine, University of Copenhagen, Copenhagen, Denmark.

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

Manuscript received July 17, 2025; revised manuscript received September 2, 2025, accepted September 5, 2025.

ABBREVIATIONS AND ACRONYMS

AF = atrial fibrillation
BMI = body mass index
ILR = implantable loop recorder
OAC = oral anticoagulant
PC = principal component
PRS = polygenic risk score
QC = quality control
SE = systemic embolism

Atrial fibrillation (AF) is common, with prevalence rising steeply with age and the accumulation of cardiovascular and cardiometabolic risk factors.^{1,2} It is a major contributor to ischemic stroke and impaired health status,^{3,4} but a substantial proportion of AF is asymptomatic or “silent,” particularly in older people. The rationale for screening is therefore straightforward: if undiagnosed AF can be detected, treatment can be initiated to prevent complications.

SEE PAGE 154

Despite technological advances, the net benefit of systematic screening remains debated.⁵ The STROKESTOP (Systematic ECG Screening for Atrial Fibrillation Among 75 Year Old Subjects in the Region of Stockholm and Halland, Sweden) trial randomized all 75-year-olds in the trial’s uptake area to invitation to intermittent electrocardiogram screening vs no invitation and showed a statistically significant reduction in a composite outcome, though the effect sizes were modest, AF detection was low, and stroke risk was not significantly reduced.⁶ The LOOP (Atrial Fibrillation Detected by Continuous ECG Monitoring Using Implantable Loop Recorder to Prevent Stroke in High-Risk Individuals) study enrolled individuals at high risk of stroke for screening with an implantable loop recorder (ILR) vs usual care.⁷ This trial demonstrated a 3-fold increase in AF diagnoses and anticoagulant use but showed no significant reduction in stroke risk, and bleeding events were numerically higher.⁷ Both trials were arguably limited by AF detected through usual care and by competing health risks unaffected by screening.

Screening is resource intensive, and any subsequent treatment carries a risk of complications. The evidence underscores that “more detection” alone does not guarantee benefit. The guidelines reflect this nuance: they endorse opportunistic pulse or electrocardiogram checks in older adults presenting for health care for other reasons, but they stop short of recommending universal, large-scale systematic screening.^{1,2} Still, screening is increasing, both in health care settings and in the community (eg, consumer-led screening). In research and daily practice, the key question remains of whether screening can be refined to target those most likely to benefit while avoiding unnecessary testing in individuals unlikely to gain from it.

With an estimated heritability of up to 22% and >300 associated genetic loci,⁸ AF has a substantial genetic component. Studies using polygenic risk

scores (PRS) have also shown potential for identifying individuals at increased risk of AF and stroke.⁹⁻¹¹ The integration of genetics therefore offers a unique and previously unexplored opportunity in AF screening.

In the current prespecified subanalysis of the LOOP study we investigated screening efficacy according to genetic risk. Here, we present the first findings from a randomized study on the effects of screening on stroke prevention, supplemented by analyses of AF detection, major bleeding events, and mortality.

METHODS

STUDY DESIGN. The present study was a pre-specified post hoc analysis of the LOOP study, a multicenter, randomized controlled trial conducted across 4 centers in Denmark. Methodology and findings from the LOOP study have previously been described in detail.^{7,12} The study included a total of 6,004 AF-naïve individuals aged 70 to 90 years, with at least 1 additional stroke risk factor (hypertension, diabetes, heart failure, or previous stroke). The participants were randomized 1:3 to receive either continuous ILR monitoring and initiation of anticoagulation upon detection of AF episodes lasting at least 6 minutes or usual care. A baseline visit included detailed medical history, vital signs, blood samples, and more. Written informed consent was obtained from all participants at enrollment. The study was approved by the Regional Scientific Ethics Committee for the Capital Region of Denmark (H-42013-025).

GENETIC SEQUENCING AND QUALITY CONTROL. Genetic sequencing was conducted by extracting genomic DNA from leukocytes in whole blood. Genotyping was performed using Infinium Global Screening Array v2 (Illumina) at the Institute of Clinical Molecular Biology (IKMB), Kiel University, Germany. Preimputation quality control (QC) was performed at the sample level to exclude individuals with >2% missing genotypes, individuals with sex mismatch, by heterozygosity rate outliers >3 SD, and related individuals (first and second degree), and to remove outliers in a principal component (PC) analysis. QC at the variant level included removal of markers with >2% missingness and a Hardy-Weinberg equilibrium filter of P value $<1 \times 10^{-10}$. Imputation was performed at the Michigan imputation server¹³ using the Haplotype Reference Consortium version r1.1 2016 as the reference.¹⁴ After imputation we applied further filtering for imputation quality ($R^2 > 0.3$) and minor allele frequency

(MAF >0.01). A flowchart of study design and genetic QC is shown in [Supplemental Figure 1](#).

POLYGENIC RISK SCORE FOR AF. Weights were obtained from the Cardiovascular Disease Knowledge Portal for a previously published PRS on AF.¹⁵ This PRS has been evaluated in several other cohorts, including the Stroke Genetics Network and the UK Biobank.^{10,15} The underlying methods of the PRS calculation have previously been described in detail.¹⁵ In brief, the PRS incorporated 1,111,128 genetic variants, with weights calculated based on a genome-wide association study on AF¹⁶ using the PRS-CS algorithm. This algorithm is based on a Bayesian approach, which is robust to varying genetic architectures.¹⁷ The PRS_{AF} and its underlying dataset are described in detail in the [Supplemental Material](#). We then used PLINK¹⁸ to compute an individual-level PRS_{AF} for each participant in the LOOP study with genetic data. The PRS_{AF} was subsequently normalized by scaling to a mean of 0 and an SD of 1. The PRS_{AF} was evaluated by comparing a baseline model adjusted for sex, age, body mass index (BMI), and 6 first genotype PCs, with a model further adjusted for PRS_{AF}. Performance was calculated for incident AF as an outcome using C-statistics.

To assess whether potential associations were driven by variants in the *PITX2* loci, we conducted sensitivity analyses using a PRS that did not include variants near this locus ([Supplemental Material](#)).

OUTCOMES. In accordance with the LOOP study, the primary outcome in the present study was a composite endpoint of stroke or systemic embolism (SE). Secondary outcomes were AF, major bleeding (as defined by the International Society on Thrombosis and Haemostasis),¹⁹ and all-cause mortality. The endpoints stroke, SE, major bleeding, and death were adjudicated by a clinical endpoint committee. ILR-detected AF was confirmed by at least 2 senior cardiologists.⁷

As an exploratory analysis we examined whether PRS_{AF} was associated with having long-lasting episodes of AF. These analyses were conducted in participants with continuous cardiac monitoring and with adjudicated AF during follow-up. We defined long-lasting AF as having ≥1 episode lasting ≥24 hours.

STATISTICAL ANALYSES. Sample size and power calculation of the LOOP study trial have previously been described elsewhere.⁷ Clinical outcomes were assessed in time-to-first-event analyses, following participants until the event of interest, death, or end of follow-up, whichever came first. Hazard ratios were calculated applying cause-specific Cox

regression models with adjustment for sex, age, BMI, estimated glomerular filtration rate, alcohol intake, smoking status, plasma cholesterol, plasma thyroid-stimulating hormone, history of hypertension, diabetes, prior stroke, valvular disease, coronary artery disease, congestive heart failure, and peripheral artery disease, and the first 6 genotype PCs. Interactions were assessed by including PRS_{AF} as an interaction term in the Cox models. HRs for ILR screening vs usual care were evaluated with the PRS_{AF} as a continuous variable by constructing a prediction grid using the *rms* package. The contrast between randomization for ILR vs usual care was calculated across the observed range of the PRS_{AF}, modeled as a linear variable in 0.1 SD increments. Results are presented as HR according to ILR vs usual care with 95% confidence intervals.

As a secondary analysis, HRs and incidence for ILR screening vs usual care according to specific genetic predisposition for AF were evaluated by stratifying the population into 2 subgroups, divided at the median of the PRS_{AF}. Crude event rates per 100-person years during follow-up were calculated using a Poisson regression. Cumulative incidences were calculated using the Kaplan-Meier method for all-cause mortality. For all other outcomes, the Aalen-Johansen estimator was used, considering all-cause death as a competing risk. Survival analyses were conducted using the *survival* and *prolim* packages in R version 4.2.0 (R Project for Statistical Computing). Cumulative incidences were presented at 6 years of follow-up. The numbers needed to screen, and harm were calculated in subgroup analyses using the absolute risks at 6 years with confidence intervals based on binomial approximation.

In another secondary analysis, we analyzed associations between PRS_{AF} and long-lasting AF episodes among individuals with detected AF on continuous cardiac monitoring using multivariable logistic regressions. The models were adjusted as described previously.

We considered a 2-tailed $P < 0.05$ as statistically significant. Covariates were selected based on clinical risk factors for AF and stroke. Potential multicollinearity was evaluated using generalized variance inflation factors, assessed as $GVIF^{1/(2DF)}$ to compare predictors with different degrees of freedom. No $GVIF^{1/(2DF)}$ values exceeded 2. The Cox proportional hazard assumption for the exposure variable was evaluated using Schoenfeld residuals and showed satisfactory consistency for all outcomes across the study period. To assess consistency across different clinical strata, we examined interactions between PRS_{AF} and age, sex, and CHA₂DS₂-VASc respectively

TABLE 1 Baseline Characteristics at Enrollment

Study Cohort ^a	Overall (N = 5,656)	Low PRS _{AF} (n = 2,828)	High PRS _{AF} (n = 2,828)
Sex			
Female	2,670 (47.2)	1,295 (45.8)	1,375 (48.6)
Male	2,986 (52.8)	1,533 (54.2)	1,453 (51.4)
Age, y	74.4 (71.6-77.0)	73.3 (71.6-77.2)	73.1 (71.6-76.8)
BMI, kg/m ²	27.7 ± 4.6	27.7 ± 4.6	27.7 ± 4.5
eGFR, mL/min	69.9 ± 15.7	70.6 ± 15.9	70.2 ± 15.8
P-cholesterol, mmol/L ^b	4.8 ± 1.1	4.8 ± 1.1	4.8 ± 1.1
Comorbidities			
Coronary artery disease	754 (13.3)	416 (14.7)	338 (12.0)
Diabetes	1,597 (28.2)	788 (27.9)	809 (28.6)
Hypertension	5,138 (90.8)	2,587 (91.5)	2,551 (90.2)
Peripheral artery disease	152 (2.7)	73 (2.6)	79 (2.8)
Stroke	986 (17.4)	481 (17.0)	505 (17.9)
Valvular heart disease	228 (4.0)	104 (3.7)	124 (4.4)
Smoking ^c			
Current	520 (9.2)	261 (9.2)	259 (9.2)
Previous	2,896 (51.2)	1,488 (52.6)	1,408 (49.8)
Never	2,239 (39.6)	1,078 (38.1)	1,161 (41.1)
Alcohol intake, drinks/wk	7.3 ± 8.1	7.5 ± 8.4	7.1 ± 7.9

Values are n (%), median (Q1-Q3), or mean ± SD. ^aBaseline characteristics at enrollment in the LOOP study, stratified by the median of atrial fibrillation polygenic risk score. ^bP-cholesterol denotes total cholesterol in plasma. ^c1 participant did not report smoking status.

AF = atrial fibrillation; BMI = body mass index; eGFR = estimated glomerular filtration rate; PRS = polygenic risk score.

for each outcome (AF, stroke/SE, major bleeding, and death).

RESULTS

Among the 6,004 participants in the LOOP study, 5,656 had available genetic data that passed QC, corresponding to 94.2% of the trial population. The median age at enrollment was 73.2 years (Q1-Q3: 71.6-77.0 years), and 47.2% were women. Baseline characteristics at enrollment are summarized in [Table 1](#). The distribution of the PRS_{AF} by randomization arm and outcome is shown in [Supplemental Figure 2](#).

AF AND OUTCOMES ACCORDING TO GENETIC RISK.

During a median follow-up period of 5.4 years (Q1-Q3: 4.9-5.2 years), 969 participants (17.1%) received a diagnosis of AF, 980 (17.3%) initiated oral anticoagulant (OAC) treatment, 296 (5.2%) had a stroke/SE, 206 (3.6%) had a major bleeding, and 624 (11.0%) died. None were lost to follow-up.

The PRS_{AF} was significantly associated with incident AF (HR per 1-SD increase: 1.20; 95% CI: 1.13-1.28; $P < 0.001$). The addition of PRS_{AF} resulted in a $\Delta = 0.015$ in C-statistics, compared with a reference model adjusted for sex, age, BMI, and 6 first genotype PCs (C-statistic: 0.601; 95% CI: 0.583-0.619 vs 0.622; 95% CI: 0.604-0.639).

A 1-SD increase in PRS_{AF} was associated with significantly increased rates of anticoagulation initiation (HR per 1-SD increase: 1.21; 95% CI: 1.14-1.29; $P < 0.001$). We observed nonsignificant trends toward higher rates of stroke/SE (HR per 1-SD increase: 1.10; 95% CI: 0.98-1.24; $P = 0.099$), major bleeding (HR per 1-SD increase: 1.11; 95% CI: 0.97-1.28; $P = 0.12$), and all-cause mortality (HR per 1-SD increase: 1.06; 95% CI: 0.98-1.15; $P = 0.17$). Cumulative incidences of AF, stroke/SE, and all-cause death are visualized in [Figure 1](#).

We identified a positive interaction between PRS_{AF} and CHA₂DS₂-VASC score for major bleeding (HR per 1-SD increase per CHA₂DS₂-VASC point: 1.12; $P = 0.026$). We did not observe any other significant interactions between PRS_{AF} and age, sex, and CHA₂DS₂-VASC score ($P_{\text{interaction}} > 0.05$).

AF AND ANTICOAGULATION INITIATION ACROSS GENETIC RISK GROUPS.

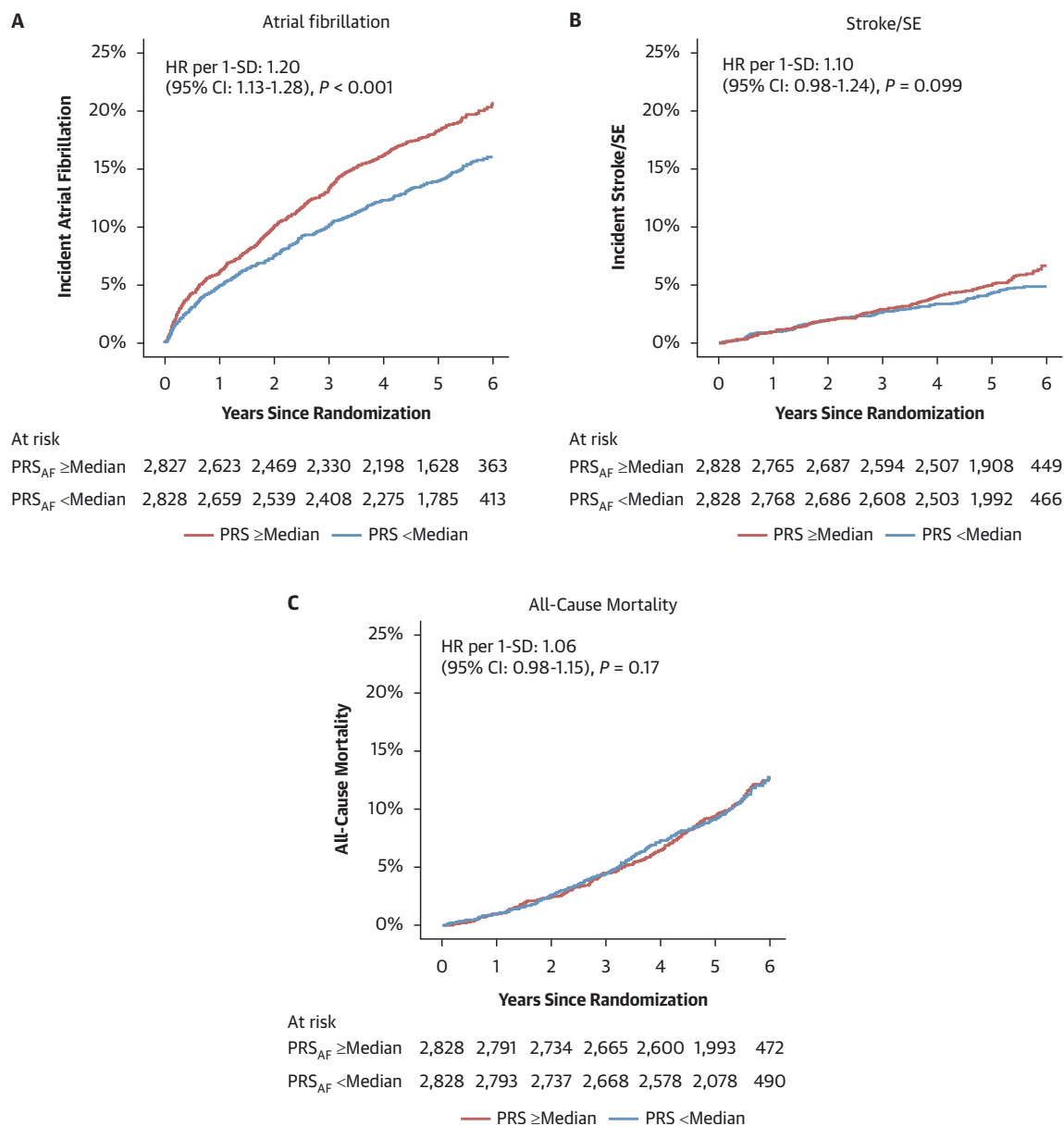
The detection of AF according to randomization and genetic risk subgroups is illustrated in [Supplemental Figure 3](#). Stratified by genetic risk, ILR screening was associated with equally increased AF detection among those with PRS_{AF} <median (HR: 3.28; 95% CI: 2.71-3.98; $P < 0.001$) and those with PRS_{AF} ≥median (HR: 3.40, 95% CI: 2.87-4.03; $P < 0.001$; $P_{\text{interaction}} = 0.31$) compared with the respective controls. Among participants diagnosed with AF, 88.8% initiated anticoagulation, with minor differences between groups. Among participants with PRS_{AF} <median, 88.4% of those randomized to usual care and 86.6% of those randomized to ILR screening initiated anticoagulation. For participants with PRS_{AF} ≥median, 86.2% those randomized to usual care and 93.7% of participants randomized to ILR-screening started anticoagulation. The number of outcomes and OAC initiation for all participants across the different strata are summarized in [Supplemental Table 1](#). Event rates of each outcome per 100 person-years are shown in [Figure 2](#).

Among 431 individuals with ILR-detected AF, 77 (17.9%) had AF episodes lasting ≥24 hours. A 1-SD increase in PRS_{AF} was significantly associated with having ≥1 long-lasting episodes (OR: 1.35; 95% CI: 1.02-1.78; $P = 0.037$).

ILR SCREENING EFFICACY ACCORDING TO GENETIC RISK.

We observed a significant interaction between ILR screening and genetic risk of AF for both stroke/SE ($P_{\text{interaction}} = 0.006$) and for major bleeding ($P_{\text{interaction}} = 0.036$). When analyzing ILR screening effects across the continuous range of the PRS_{AF} ([Figure 2](#)), the trend curve crossed the reference line (HR: 1.00) for stroke/SE at -0.51 SD below the median

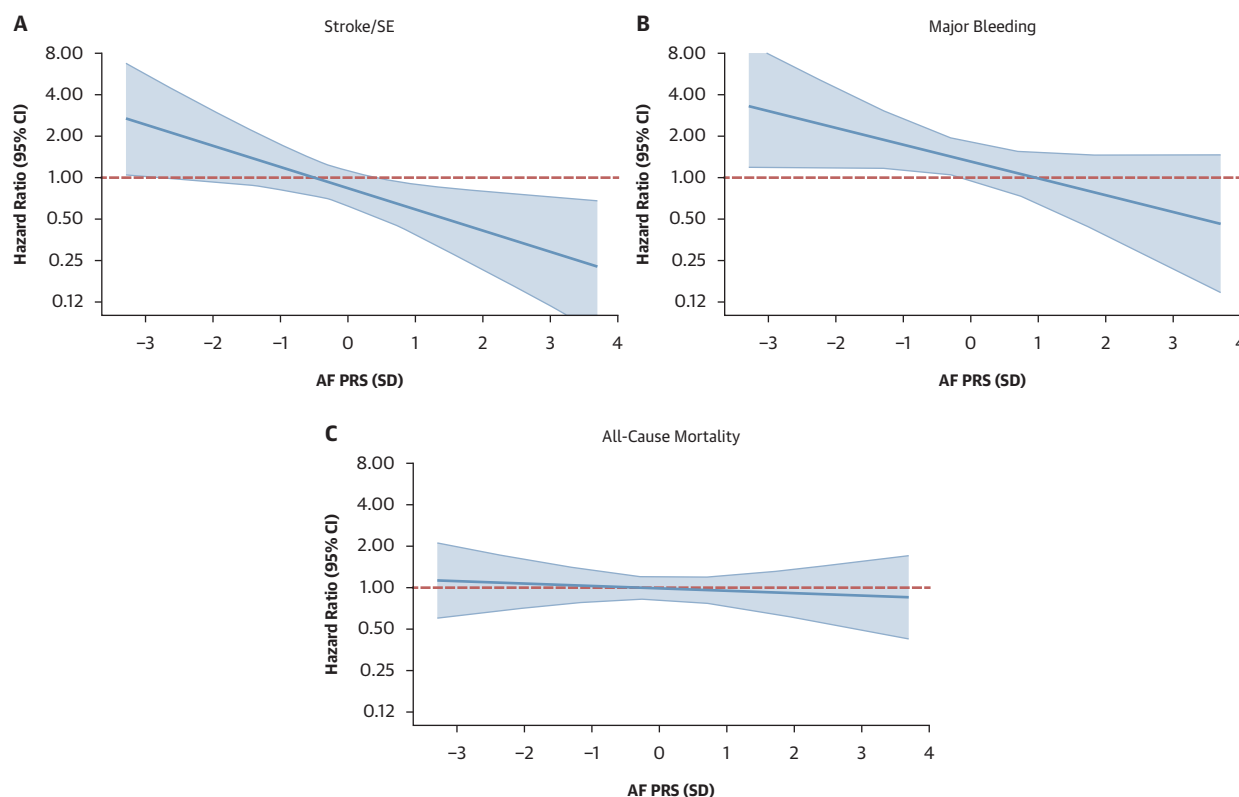
FIGURE 1 Rates of Atrial Fibrillation, Stroke, and Death According to Genetic Risk



Cumulative incidences of (A) atrial fibrillation, (B) stroke/SE, and (C) all-cause mortality stratified by genetic risk of atrial fibrillation. Participants with PRSAF ≥ median are shown in red, and participants with PRSAF < median are shown in blue. AF = atrial fibrillation; ILR = implantable loop recorder; PRS = polygenic risk score; SE = systemic embolism.

PRSAF, with increasing efficacy of ILR screening on prevention of stroke/SE above this point. For major bleeding in individuals randomized to ILR screening, the trend curves crossed the reference line at 0.95 SD above the mean PRSAF, with higher rates of major bleeding in participants with PRSAF below this cutoff.

For the primary outcome of stroke or SE, ILR screening was associated with lower rates in individuals with PRSAF ≥ median (HR: 0.65; 95% CI: 0.43-0.97; $P = 0.036$), but not among individuals with PRSAF < median (Figure 3). Among participants with PRSAF ≥ median, this corresponded to a cumulative

FIGURE 2 Screening Efficacy Across Continuous Polygenic Risk Score Range

HRs for atrial fibrillation, stroke/SE, and all-cause mortality stratified by randomization to ILR-screening or usual care, across the genetic risk spectrum (x-axis). Colored areas denote 95% confidence intervals. Abbreviations as in [Figure 1](#).

incidence of stroke/SE of 7.39% (95% CI: 6.03%-8.76%) among those randomized to usual care, and those randomized to ILR screening exhibited an absolute risk of 4.41% (95% CI: 2.82%-6.01%) ([Figures 4A and 4B](#)). Given the anticipated absolute stroke risk reduction of 2.98% (ie, 7.39-4.41), the number needed to screen to avoid 1 primary outcome after 6 years would be 34 (95% CI: 20.6-90.8) among participants with $PRS_{AF} \geq \text{median}$. By comparison, absolute risks of major bleeding in this subgroup were 4.44% (95% CI: 2.78%-6.10%) for participants randomized to ILR and 4.32% (95% CI: 3.36%-5.28%) for participants randomized to usual care, corresponding to a difference in absolute risk of 0.12% for major bleeding.

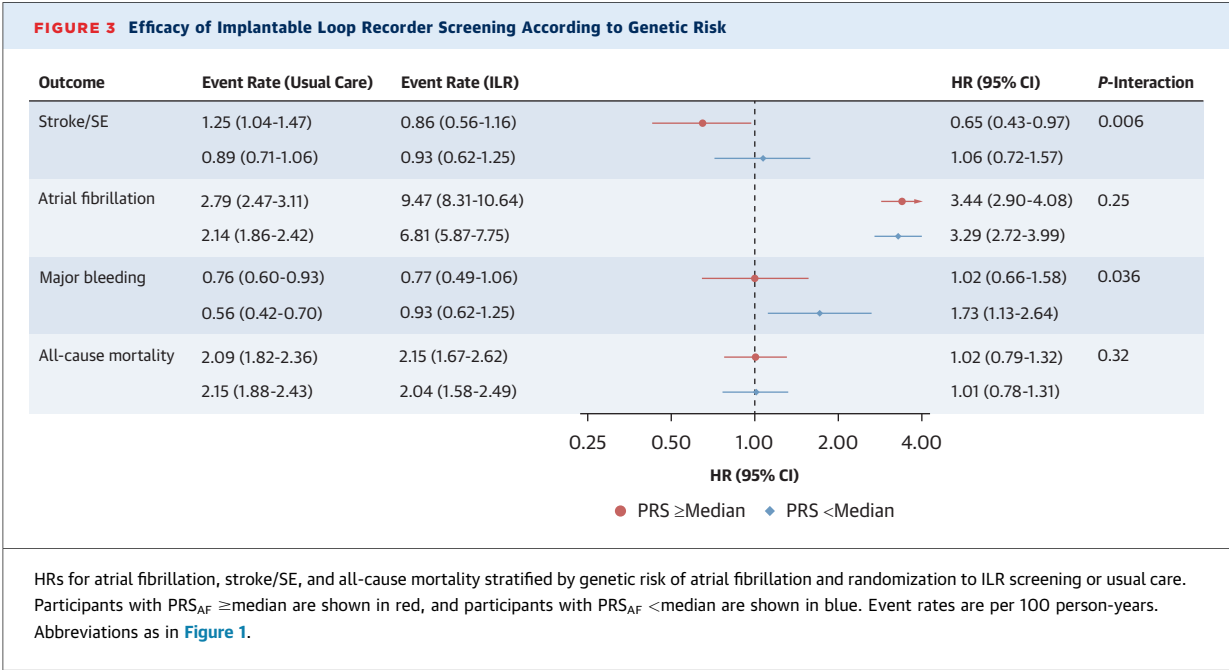
By contrast, ILR screening was associated with higher rates of major bleeding in individuals with $PRS_{AF} < \text{median}$ (HR: 1.71; 95% CI: 1.12-2.64; $P = 0.011$) but not in individuals with $PRS_{AF} \geq \text{median}$ ([Figure 3](#)). Among participants with $PRS_{AF} < \text{median}$, those randomized to usual care had an absolute risk

of major bleeding of 2.92% (95% CI: 2.18%-3.32%). By comparison, those randomized to ILR screening showed an absolute risk of 5.43% (95% CI: 3.52%-7.34%) for major bleeding ([Figures 4C and 4D](#)). Given the absolute risk increase of major bleeding at 6 years of 2.51% (ie, 5.43%-2.92%), the number needed to harm among participants with $PRS_{AF} < \text{median}$ would be 40 (95% CI: 23.2-142.0).

We found no evidence of an interaction between ILR screening and PRS_{AF} for all-cause mortality ($P_{\text{interaction}} = 0.31$) and observed no significant association between ILR screening and death in either PRS_{AF} group ([Figure 3](#)).

DISCUSSION

The present study is the first to evaluate the potential efficacy of genetically guided AF screening to prevent clinical outcomes and, to our knowledge, the first post hoc analysis of a nononcology screening randomized controlled trial to demonstrate that genetic



susceptibility, measured as a PRS, may benefit screening selection. Although previous studies have shown that genetic predisposition can help identify individuals at risk of AF,⁹ evidence is lacking that the pursuit of AF detection can reduce stroke. In other words, screen-detected AF may not carry the same risk as clinically diagnosed AF, and not all AF may be worth screening for or warrant anticoagulation. In this post hoc analysis of a large, randomized trial, we sought to address this knowledge gap.

Our findings suggest that a PRS for AF can potentially be used to select individuals who benefit from screening and anticoagulation for screen-detected AF. We found that individuals with high genetic predisposition for AF had a significant reduction in stroke when screened for AF. By contrast, those with low genetic predisposition had no benefit from screening despite having multiple clinical risk factors. Instead, screening and subsequent anticoagulation in these individuals was associated with increased rates of bleeding without any reduction in strokes.

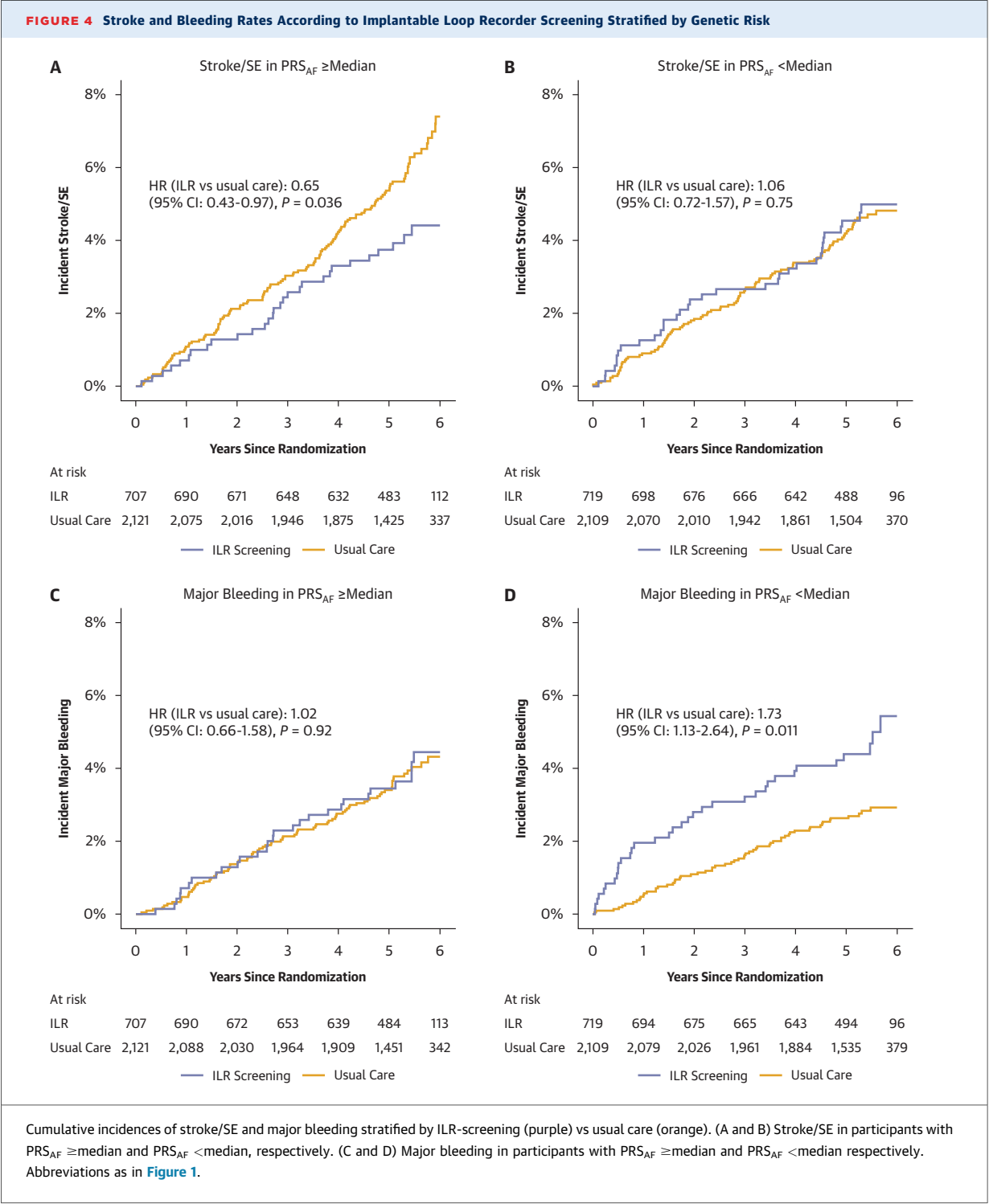
Previous studies have shown that genetic predisposition can aid in identifying high-risk individuals for clinical intervention. This includes 2 recent studies showing a greater benefit of cholesterol-lowering therapies in individuals with a high PRS for coronary artery disease.^{20,21} Notably, the interventions also benefited individuals at lower genetic risk. In a post hoc study of the EAST-AFNET 4 (Treatment of AF for Stroke Prevention Trial), early rhythm control was shown to be effective across the genetic risk

spectrum.²² By contrast, our findings did not show a benefit across the entire genetic spectrum but instead identified specific groups who do and do not benefit.

PRS TO SELECT INDIVIDUALS WITH BENEFIT FROM SCREENING. Compared with the previously reported 62 persons needed to screen to avoid 1 primary outcome from the LOOP study,⁷ our findings indicate that this could be reduced to 34 persons if the study had selected participants based on genetic risk. This may potentially reflect that those at a higher genetic risk are more likely to have clinically relevant AF phenotypes with higher likelihood of benefit if diagnosed and treated. While this notion, and its underlying biological explanation warrant further study, an exploratory analysis among participants with ILR-detected AF showed that a higher PRS_{AF} was associated with a higher OR having at least 1 AF episode lasting ≥24 hours. These results hint that individuals with a higher genetic risk may be at a greater predisposition for higher AF burden.

Cost-effectiveness and potential harm from unnecessary treatment represent 2 important concerns when evaluating the potential benefits of screening programs. Our findings illustrate that using genetic risk to refine the selection of participants for screening may help by reducing the number needed to screen and by identifying persons who would not benefit. Specifically, our findings suggested that restricting screening to individuals of higher genetic risk was associated with an absolute risk reduction of

FIGURE 4 Stroke and Bleeding Rates According to Implantable Loop Recorder Screening Stratified by Genetic Risk



2.98% compared to an increase in risk of major bleeding of 0.12%, and the relative risk of major bleeding among those randomized to ILR screening was higher among participants with low PRS_{AF}. This appeared primarily to be driven by higher rates of major bleeding in the usual care arm among individuals with high PRS_{AF} because major bleeding rates were relatively similar among participants

randomized to ILR screening in both genetic risk groups. We note that our finding of increased bleeding risk after screening compared with usual care in those at low genetic risk is more fragile than our finding of stroke prevention in those at high genetic risk.

Although our findings should be regarded as hypothesis-generating in nature, the higher rates of both stroke and bleeding among participants with PRS_{AF} above rather than below median within the usual care arm may reflect a higher health risk overall in these individuals. Interestingly, we observed a modest positive interaction between PRS_{AF} and CHA₂DS₂-VASC score for major bleeding. However, given the high baseline risk of the cohort and multiple interaction tests, this finding should be interpreted cautiously.

The findings should be viewed in the context of other screening-by-subgroup interaction analyses conducted in the LOOP study. Notably, with the exception of an interaction with systolic blood pressure, no screening-by-subgroup interactions were identified across a variety of clinical variables, including CHA₂DS₂-VASC score and age. This may be due to the enrollment criteria in the trial, where all participants were aged ≥ 70 years and had CHA₂DS₂-VASC scores ≥ 2 with a median of 4. In such a high-risk population, the integration of genetic data may offer incremental benefits in the selection who would benefit from systematic screening. A related benefit from screening would arguably be implausible in a low-risk population without indications for anticoagulation.

FUTURE DIRECTIONS. Genetic risk to select individuals for screening is being employed in cancer research and has shown promising preliminary results in screening for prostate cancer²³ and breast cancer.²⁴ Our findings expand the field, indicating that the use of genetics in screening could be applied to a broader range of disorders, including cardiovascular diseases. The ongoing GenoVA (Genomic Medicine at Veterans Affairs; [NCT04331535](#)) study,²⁵ an ambitious clinical trial, is currently investigating the implementation of PRS in primary care, examining time to disease, diagnostic testing, and medication adherence across various outcomes. AF is among the diseases investigated, where earlier diagnosis and rhythm control could potentially lower the risk of adverse cardiovascular outcomes.²⁶ Although there is much potential to explore clinical applications of PRS with the GenoVA study, we note that medical treatment is not per se decided based on PRS.

For AF, large-scale studies based on data from the AFGen consortium and biobanks like the UK Biobank²⁷ and All of Us²⁸ have elucidated a complex genetic landscape, with the latest genome-wide association study reporting >300 genetic loci associated with AF.⁸ Although these findings form the foundation for our understanding of the genetic underpinnings of AF and the construction of corresponding PRS, the underlying populations are not necessarily ideal for identifying how genetic risk might be implemented in clinical practice. For instance, the participants in biobanks such as the UK Biobank are often healthier and more socioeconomically privileged than the background population.²⁹ By contrast, the implementation of genetic predisposition as a risk stratifier would likely be more relevant in high-risk population groups, and its use would need to be evaluated in the setting of a randomized clinical trial.

At the present, findings on genetic data from current trials on AF screening may provide insights into what might be expected from future trials on genetically guided AF screening. Among the AF-screening trials, the STROKESTOP trial reported a small benefit of screening⁶ but has not reported genetic findings. The EAST-AFNET trial recently reported genetic findings on a subset of 1,567 participants and identified no significant interactions between PRS and early rhythm control. Thus, the results of our present study represent the largest genetic dataset in an AF-screening trial as well as the first insights into a potential interaction between genetic risk and AF-screening efficacy.

As genetic sequencing technology advances and becomes more available, the implementation of genetic information in clinical settings is likely to play a larger role in the future. The use of genetics for screening and risk stratification is addressed in AF guidelines and consensus statements from both the American College of Cardiology and the American Heart Association² and the European Society of Cardiology.^{1,30} However, both American and European societies stress that there are still considerable knowledge gaps that hinder the implementation of PRS in clinical practice, including a lack of evidence from clinical trials. Our findings further the understanding of potential applications of genetics in cardiovascular disease, but more large-scale clinical trials are needed. We advocate for continued exploration of genetics in cardiovascular disease prevention, including the possibility of taking PRS into consideration when deciding on screening or medical treatment to prevent disease progression and complications.

Based on the present studies, one could envision a future trial where participants first undergo genetic risk scoring and only those with an expected benefit from screening (per current analyses) are randomized to screening vs usual care. The limitations would include dropout, because some with high genetic risk might decline screening, and a potential Hawthorne effect because individuals informed of their high risk may independently seek screening. An elegant alternative would be for a health system to identify high-risk individuals directly—such as from previously collected samples with consent for research—and randomize this cohort to receive an invitation to screening vs no invitation.

STUDY LIMITATIONS. The present study should be interpreted with respect to the study limitations. First, the study population consisted of individuals aged 70 to 90 years with additional risk factors for stroke. Although this is a strength of our study design because it allows us to examine stroke prevention in the part of the population at the highest risk, it also represents a limitation concerning generalizability to the broader population. In this context, we note that most large-scale genome-wide association studies, including the one underlying the PRS_{AF} , reflect lifetime risk of AF and not the risk of incident AF in a clinical trial setting. Therefore, the LOOP study cohort may in some respects differ in composition with regards to age and comorbidities compared with the population underlying the genetic data of the PRS_{AF} . Furthermore, although this post hoc study was prespecified, post hoc results should, naturally, be interpreted with caution, and we note that our study may have been underpowered to accurately delineate screening efficacy across all genetic risk groups. In this context, it is important to acknowledge that the statistical analysis plan was written for the primary clinical trial, and subsequent analyses, even if predefined as is the case in the present study, should be regarded as hypothesis generating.

It is important to acknowledge that although the hypothesized crossing point found for PRS_{AF} may aid in visualizing how participants could be stratified for screening, this analysis was exploratory, and potential formal cutoff points need to be investigated in future clinical trials. We also note that analyses of major bleeding events and AF burden should be regarded as suggestive. As a multicenter study in Denmark, the population was of European ancestry, and the PRS_{AF} used was primarily based on genetic findings from European ancestry individuals. This may limit generalizability to other populations. The study population was derived from a randomized

clinical trial, but we cannot exclude some degree of residual confounding.

CONCLUSIONS

This post hoc analysis of the randomized LOOP study is the first study to examine genetic risk as a potential tool to guide screening for AF to prevent stroke. Compared with usual care, we found that continuous cardiac monitoring followed by OAC initiation upon AF detection was associated with lower stroke risk in individuals with a high PRS for AF. By contrast, individuals with lower genetic risk did not benefit from ILR screening but presented with higher rates of major bleeding. These exploratory findings suggest that genetic risk profiling may support early AF detection strategies for stroke prevention.

DATA AVAILABILITY Due to privacy and ethical concerns, individual-level data from the LOOP study cannot be shared. Code and methodology will be shared upon reasonable request to the corresponding author (SZD).

ACKNOWLEDGMENTS The authors thank Dan Atar (University of Oslo and Department of Cardiology, Oslo University Hospital Ullevål, Oslo, Norway) and Mårten Rosenqvist (Karolinska Institutet and Danderyd Hospital, Stockholm, Sweden) for assisting the study with their expertise in the study's international advisory committee. They also thank the research nurses and other colleagues in the Departments of Cardiology at Copenhagen University Hospital-Rigshospitalet, Copenhagen University Hospital-Bispebjerg, and Frederiksberg, Zealand University Hospital-Roskilde, and Odense University Hospital who assisted with study procedures.

FUNDING SUPPORT AND AUTHOR DISCLOSURES

The LOOP Study was supported by Innovation Fund Denmark (12-135225), the Research Foundation for the Capital Region of Denmark, the Danish Heart Foundation (11-04-R83-A3363-22625), Aalborg University Talent Management Program, Arvid Nilssons Fond, Skibsrøder Per Henriksen, R og Hustrus Fond, the AFFECT-EU Consortium (Horizon 2020, grant number 847770-AFFECT-EU), Læge Sophus Carl Emil Friis og hustru Olga Doris Friis' Legat, and an unrestricted grant from Medtronic. Dr Vad was funded by the Research Foundation of Rigshospitalet; and was mentored through the AFGen fellowship for early-career researchers 2024-2025 (funded by AFGen grant R01HL092577). The funders of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report. Dr Diederichsen is a part-time employee of Vital Beats; has received consultancy fees from Cortrium, Acesion Pharma, Bristol Myers Squibb-Pfizer, and Novo Nordisk; has received lecture fees from Bayer, Bristol Myers Squibb-Pfizer, and AstraZeneca; and has received travel grants from Abbott and Boston Scientific. Dr Brandes has received lecture fees from Bristol Myers Squibb; and has received

research grants from Theravance, the Danish Heart Foundation, the European Union Interreg 5A Programme, and the Independent Research Fund Denmark. Dr Svendsen is a member of the advisory board for Vital Beats and Medtronic; and has received speaker honoraria and research grants from Medtronic in relation to this work and outside this work. All other authors have reported that they have no relationships relevant to the contents of this paper to disclose.

ADDRESS FOR CORRESPONDENCE: Dr Søren Z. Diederichsen, Department of Cardiology, Copenhagen University Hospital-Rigshospitalet, Inge Lehmanns Vej 7, Copenhagen, Copenhagen DK-2100, Denmark. E-mail: soeren.zoega.diederichsen@regionh.dk.

REFERENCES

1. Van Gelder IC, Rienstra M, Bunting KV, et al. 2024 ESC guidelines for the management of atrial fibrillation developed in collaboration with the European Association for Cardio-Thoracic Surgery (EACTS): developed by the Task Force for the Management of Atrial Fibrillation of the European Society of Cardiology (ESC), with the special contribution of the European Heart Rhythm Association (EHRA) of the ESC. Endorsed by the European Stroke Organisation (ESO). *Eur Heart J*. 2024;45:3314–3414.
2. Joglar JA, Chung MK, Armbruster AL, et al. 2023 ACC/AHA/ACCP/HRS guideline for the diagnosis and management of atrial fibrillation: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *J Am Coll Cardiol*. 2024;83(1):109–279.
3. Vinter N, Cordsen P, Johnsen SP, et al. Temporal trends in lifetime risks of atrial fibrillation and its complications between 2000 and 2022: Danish, nationwide, population based cohort study. *BMJ*. 2024;385:e077209.
4. Paludan-Müller C, Vad OB, Stampe NK, et al. Atrial fibrillation: age at diagnosis, incident cardiovascular events, and mortality. *Eur Heart J*. 2024;45(24):2119–2129.
5. Xing LY, Vad OB, Engler D, Svendsen JH, Diederichsen SZ. Screening for atrial fibrillation: the role of CHA2DS2-VASc and atrial fibrillation burden. *Eur Heart J Suppl*. 2024;26:iv41–iv49.
6. Svennberg E, Friberg L, Frykman V, Al-Khalili F, Engdahl J, Rosenqvist M. Clinical outcomes in systematic screening for atrial fibrillation (STROKESTOP): a multicentre, parallel group, unmasked, randomised controlled trial. *Lancet*. 2021;398:1498–1506.
7. Svendsen JH, Diederichsen SZ, Højberg S, et al. Implantable loop recorder detection of atrial fibrillation to prevent stroke (the LOOP study): a randomised controlled trial. *Lancet*. 2021;398:1507–1516.
8. Roselli C, Surakka I, Olesen MS, et al. Meta-analysis of genome-wide associations and polygenic risk prediction for atrial fibrillation in more than 180,000 cases. *Nat Genet*. 2025;57:539–547.
9. Marston NA, Garfinkel AC, Kamanu FK, et al. A polygenic risk score predicts atrial fibrillation in cardiovascular disease. *Eur Heart J*. 2023;44:221–231.
10. Vad OB, Monfort LM, Paludan-Müller C, et al. Rare and common genetic variation underlying atrial fibrillation risk. *JAMA Cardiol*. 2024;9(8):732–740.
11. Mishra A, Malik R, Hachiya T, et al. Stroke genetics informs drug discovery and risk prediction across ancestries. *Nature*. 2022;611:115–123.
12. Diederichsen SZ, Haugan KJ, Køber L, et al. Atrial fibrillation detected by continuous electrocardiographic monitoring using implantable loop recorder to prevent stroke in individuals at risk (the LOOP study): rationale and design of a large randomized controlled trial. *Am Heart J*. 2017;187:122–132.
13. Das S, Forer L, Schönerr S, et al. Next-generation genotype imputation service and methods. *Nat Genet*. 2016;48:1284–1287.
14. McCarthy S, Das S, Kretschmar W, et al. A reference panel of 64,976 haplotypes for genotype imputation. *Nat Genet*. 2016;48:1279–1283.
15. Weng L-C, Khurshid S, Gunn S, et al. Clinical and genetic atrial fibrillation risk and discrimination of cardioembolic from noncardioembolic stroke. *Stroke*. 2023;54:1777–1785.
16. Christophersen IE, Rienstra M, Roselli C, et al. Large-scale analyses of common and rare variants identify 12 new loci associated with atrial fibrillation. *Nat Genet*. 2017;49:946–952.
17. Ge T, Chen C-Y, Ni Y, Feng Y-CA, Smoller JW. Polygenic prediction via Bayesian regression and continuous shrinkage priors. *Nat Commun*. 2019;10:1776.
18. Purcell S, Neale B, Todd-Brown K, et al. PLINK: a tool set for whole-genome association and population-based linkage analyses. *Am J Hum Genet*. 2007;81:559–575.
19. Schulman S, Kearon C. Definition of major bleeding in clinical investigations of antithrombotic medicinal products in non-surgical patients. *J Thromb Haemost*. 2005;3:692–694.
20. Damask A, Steg PG, Schwartz GG, et al. Patients with high genome-wide polygenic risk scores for coronary artery disease may receive greater clinical benefit from alirocumab treatment in the ODYSSEY OUTCOMES trial. *Circulation*. 2020;141:624–636.
21. Natarajan P, Young R, Stitzel NO, et al. Polygenic risk score identifies subgroup with higher burden of atherosclerosis and greater relative benefit from statin therapy in the primary prevention setting. *Circulation*. 2017;135:2091–2101.
22. Kany S, Al-Taie C, Roselli C, et al. Association of genetic risk and outcomes in patients with atrial fibrillation: interactions with early rhythm control in the EAST-AFNET4 trial. *Cardiovasc Res*. 2023;119:1799–1810.
23. McHugh JK, Bancroft EK, Saunders E, et al. Assessment of a polygenic risk score in screening for prostate cancer. *N Engl J Med*. 2025;392:1406–1417.
24. Pashayan N, Morris S, Gilbert FJ, Pharoah PDP. Cost-effectiveness and benefit-to-harm ratio of risk-stratified screening for breast cancer: a life-table model. *JAMA Oncol*. 2018;4:1504–1510.
25. Vassy JL, Brunette CA, Lebo MS, et al. The GenoVA study: equitable implementation of a pragmatic randomized trial of polygenic-risk scoring in primary care. *Am J Hum Genet*. 2023;110:1841–1852.
26. Kirchhof P, Camm AJ, Goette A, et al. Early rhythm-control therapy in patients with atrial fibrillation. *N Engl J Med*. 2020;383:1305–1316.
27. Bycroft C, Freeman C, Petkova D, et al. The UK Biobank resource with deep phenotyping and genomic data. *Nature*. 2018;562:203–209.
28. Bick AG, Metcalf GA, Mayo KR, et al. Genomic data in the All of Us Research Program. *Nature*. 2024;627:340–346.
29. Fry A, Littlejohns TJ, Sudlow C, et al. Comparison of sociodemographic and health-related characteristics of UK Biobank participants with those of the general population. *Am J Epidemiol*. 2017;186:1026–1034.
30. Schunkert H, Di Angelantonio E, Inouye M, et al. Clinical utility and implementation of polygenic risk scores for predicting cardiovascular disease: a clinical consensus statement of the ESC Council on Cardiovascular Genomics, the ESC Cardiovascular Risk Collaboration, and the European Association of Preventive Cardiology. *Eur Heart J*. 2025;46(15):1372–1383.

KEY WORDS atrial fibrillation, genetics, PRS, screening, stroke

APPENDIX For supplemental material, figures, and tables, please see the online version of this paper.